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Infinite-Horizon Diversity

Why a fixed prompt has a finite novelty budget, and what to spend instead

Abstract

We study the problem of generating a corpus that stays diverse as it grows without bound. The objective is a convex combination of two terms evaluated against everything generated so far: the mean distance from a new item's embedding to the existing corpus, which we want small because it keeps items typical and on-manifold, and the minimum distance to the existing corpus, which we want large because it keeps items non-redundant. We call the resulting problem *infinite-horizon diversity* (IHD), and we are interested specifically in its asymptotics: what happens to the objective, and to the cost of satisfying it, as n grows.

Our central claim is that the interesting structure is not in the objective but in the **support**. Conditioned on a fixed prompt x , a language model's output distribution concentrates on a sub-manifold of dimension m that is far smaller than the dimension d of the reachable semantic manifold. We prove and numerically verify five consequences of that gap. Novelty at fixed prompt decays like $n^{-1/m}$, not $n^{-1/d}$, so saturation arrives much sooner than the ambient dimension suggests (measured slopes -0.472 , -0.309 , -0.195 for $m = 2, 3, 5$ against predictions -0.5 , -0.333 , -0.2). A single prompt ε -covers a vanishing εd^{-m} fraction of the manifold. Only prompt motion *transverse* to the already-occupied span raises the reachable dimension: parallel motion leaves the union's effective dimension at 1.9, transverse motion takes it to 11.2. Under free prompt-switching the optimal number of samples per prompt is exactly one; an interior optimum appears only once switching has overhead ($n^* = 10$ at cost $c = 3$, $n^* = 30$ at $c = 30$). And unconstrained max-min selection is not consistent: it picks an off-manifold candidate essentially whenever one appears in the pool, an $8.4\times$ amplification over the base rate.

We assume an embedding oracle mapping text to a fixed-size vector, and — critically — **no**

reverse oracle. We can compute exactly where we would like the next item to land and still have no way to decode that point into text. Every architectural choice follows from that asymmetry: the system must propose, measure, and select rather than solve, and the only steering handles are language-valued.

We therefore build the generation stack around latent behaviors elicited from the generator itself, organized by a **calculus of diversity** that ranks candidate latent variables by the product of four measurable factors — spread, transversality, independence, and headroom — and selects each variable's most-different values by farthest-point search in level space. When the calculus reports that no available variable clears a promise floor, it emits a *refine* decision, and the generator is asked to split the exhausted variable into finer sub-variables. The recursion is the only mechanism in the paper that changes the asymptote rather than the constant.

We test this on **43,000 real generations** from `openai/gpt-5.6-luna` across three domains — DALL·E instructions for post-modern artworks, psychometric exam items, and instrumental-music prompts — measuring diversity at three levels: literal (n -gram), latent (text embedding), and, where the artifact is not text, in the space the product actually occupies (CLIP for rendered images, CLAP and MERT for rendered audio). Four results stand out. Asked the same reasonable question ten thousand times, the model returns a psychometric corpus that is **72.3% exact duplicates**, one item repeated 1,637 times; raising the temperature to 1.6 leaves it at 61.6%, while latent conditioning drops it to 0.2%. Literal and latent diversity move in **opposite directions** as the corpus grows. Text-embedding similarity predicts *rendered-image* similarity at only $r = 0.155$, so every text-side diversity method — ours included — optimizes a proxy explaining ~2% of the variance in what the user receives. And against five real published methods (Self-Instruct, Evol-Instruct, Persona-Hub, high-temperature, naive) our method wins every metric, beating the strongest by 36% in mean-centered Vendi; against **human-written MMLU** it wins on lexical diversity and self-repetition while reaching 69% of a human exam bank's semantic diversity.

In simulation, a fixed prompt in a world with $d = 24$ and $m = 3$ reaches a Vendi Score of 1.06 where the full stack reaches 11.23, and recursive refinement lifts the reachable dimension from 13 to 24. We report two negative results that changed the design: naive refinement of saturated regions *reduces* novelty headroom, and our breadth-versus-depth theorem is false as originally stated.

1. Introduction

Ask a language model for a poem ten times and you get ten poems. Ask it ten thousand times and you do not get ten thousand poems; you get a few hundred poems and a great deal of paraphrase. This is not a defect of any particular model — it is what sampling from a fixed conditional distribution does. The distribution has a shape, and repeated sampling traces that shape ever more densely rather than expanding it.

The practical version of this problem is now everywhere. Synthetic training data is valuable in proportion to how much it is *not* already in the training set. Evaluation suites that cluster in a few families give false assurance. Test-item banks, red-team prompt sets, persona corpora, and augmentation pipelines all consume generated text in bulk, and all of them degrade in a way that standard quality metrics do not detect: every individual item is fine, and the collection is

redundant.

The natural formalization is a two-term objective. Let E be an embedding oracle and let $X_n = \{x_1, \dots, x_n\}$ be the corpus so far. For a candidate c ,

$$J_\lambda(c \mid X_n) = \underbrace{\lambda \cdot \frac{1}{n} \sum_{i=1}^n \|E(c) - E(x_i)\|}_{\text{anchor: keep it typical}} - \underbrace{(1 - \lambda) \cdot \min_{i \leq n} \|E(c) - E(x_i)\|}_{\text{repulsion: keep it new}}$$

and we accept the candidate minimizing $J\lambda$. The anchor term keeps generation on the manifold; the repulsion term pushes it away from what already exists. The interesting question is what happens to this as $n \rightarrow \infty$.

Our answer is that the objective is not where the difficulty lives. Both terms are cheap to maintain online and both behave predictably. The difficulty is that **the set of things the generator can propose is much smaller than it appears**, and the objective is powerless to enlarge it. Selection can only choose among what is offered. If the offering is confined to a thin slice, no selection rule — however clever, however well-tuned its λ — can produce diversity that the proposal distribution does not contain. Worse, as we show, aggressive selection under a fixed proposal distribution systematically degrades quality, because the only way to keep winning a novelty contest inside an exhausted region is to leave the manifold.

1.1 The oracle asymmetry

We assume throughout:

Assumption (embedding oracle, no inverse). We have access to E : $\text{Text} \rightarrow D$, computable on demand. We have **no** E^{-1} : $D \rightarrow \text{Text}$.

This is the structural fact that shapes the entire design, and it is worth dwelling on because it is easy to forget when writing down objectives. Given X_n we can compute, exactly and cheaply, the point in D that would most improve any of our diversity measures — the direction of least occupied spectral energy, the centre of the largest empty ball, the point maximizing marginal coverage. Knowing that point is worth nothing on its own. There is no procedure that turns a target embedding back into a poem.

Consequently the system cannot *solve* for its next item. It can only:

1. **propose** — sample from $p(\cdot \mid x)$ for some prompt x we are able to write;
2. **measure** — embed the proposals and score them;
3. **select** — keep one.

All of our leverage is therefore in step 1, in the choice of x , and that choice is expressible only in language. This is why the latent variables in our system are *language-valued* — named axes with named levels — rather than continuous codes: they are the only handles that reach the generator. It is also why the calculus of §4 exists. **The calculus is a surrogate for the missing inverse.** Since we cannot decode the direction we want to travel, we instead score the language-valued conditions we *can* write by how nearly their induced output distributions point that way.

1.2 Contributions

- **A conditional-dimension theory of generative saturation** (§3). Five theorems relating the conditional support dimension m to the reachable manifold dimension d , each verified numerically (`verify_slices.py`, 7/7 checks pass). The practically important one is that saturation is governed by m , which is small, and not by d .
 - **A calculus of diversity** (§4). A decision procedure for which latent variable to condition on next and which of its values to use, together with a principled termination/recursion signal. Includes an estimator bug we consider instructive enough to document (§4.3).
 - **A budget-matched evaluation protocol** (§5). Every method spends exactly 10,000 generator calls. We argue this is the only fair comparison for selection-based diversity methods, which buy diversity by discarding.
 - **Two domains with opposite diversity semantics** (§6): poetry (soft objective) and certification exam items (hard floor). The hard-floor domain exposes a failure — saturated banks silently fill with junk — that the soft domain hides.
 - **Live pilots** on `openai/gpt-5.6-luna` with locally-computed embeddings (§7).
 - **Two negative results** (§8) that we kept rather than tuned away.
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2. Related work

Diversity metrics. The Vendi Score (Friedman & Dieng, 2022) is the exponential of the von Neumann entropy of a normalized similarity matrix, interpretable as an effective number of distinct items. We use it throughout, with a computational note: under a linear kernel on unit-normalized embeddings, the $n \times n$ Gram matrix and the $D \times D$ second-moment matrix share their nonzero spectrum, so the score costs $O(nD^2 + D^3)$ rather than $O(n^3)$. We verify this to machine precision (check T5, max eigenvalue mismatch 1.1×10^{-16}). This matters for our setting specifically: an infinite-horizon method needs an evaluation whose cost does not grow super-linearly in the thing it is evaluating.

Selection and subset choice. Determinantal point processes model repulsion via volume in feature space. Farthest-point / k -center greedy is the classical max-min packing heuristic and we use it as a baseline (`fps_post`) and, at the level of axis values, as a component (§4.4). SemDeDup-style near-duplicate removal at a fixed radius appears as `dedup_post`. All of these are *post hoc* selectors over a fixed pool, which is exactly their limitation in our setting: they cannot change what is in the pool.

Quality-diversity. MAP-Elites and its descendants maintain an archive indexed by hand-designed behavior descriptors, seeking an elite per cell. Our axis lattice is a close relative, with two differences that matter here: the descriptors are *elicited from the generator rather than designed by us*, and the lattice *refines itself* when a cell saturates, so the archive's resolution is not fixed in advance.

Mode collapse in instruction-tuned models. Repeated sampling from aligned models is well documented to produce low lexical and semantic entropy relative to base models. We take this as our empirical starting point and model it explicitly as skewed mixture weights over style modes.

Novelty search. Novelty search in evolutionary computation rewards behavioral distance from

an archive. Our repulsion term is a direct analogue, and our §3.5 result is a formal statement of a hazard that community knows well in practice: unconstrained novelty pressure discovers degenerate behaviors, because degeneracy is genuinely novel.

Coverage. A companion paper treats the dual problem — maximizing the volume of a union of ε -balls at a *finite* budget, which is a covering rather than packing objective and is submodular where ours is not. §9 relates the two.

3. Theory: conditional dimension governs saturation

3.1 Setup

Let $M \subseteq D$ be the reachable semantic manifold, $\dim M = d$: the set of embeddings of texts the generator could produce under *some* prompt. For a fixed prompt x , let $S_x \subseteq M$ be the support of $E\#p(\cdot \mid x)$, with $\dim S_x = m$.

The empirical claim behind everything below is that $m \ll d$. A prompt fixes topic, stance, form, register, and rhetorical strategy; what remains free is a low-dimensional residue — some lexical choice, some ordering, some surface variation. Our live pilot supports this directly: 60 poems generated from an identical naive prompt have a median nearest-neighbour cosine similarity of 0.911 and a Vendi Score of 2.55, meaning sixty samples behave like roughly two and a half distinct items.

Throughout, $B(x, r)$ is the ball of radius r and we write gn for the min-gap of a fresh draw against a corpus of size n .

3.2 Fixed-prompt saturation

Theorem 1. Let X_n be n i.i.d. draws from a distribution with density bounded above and below on an m -dimensional manifold S . For a fresh draw c from the same distribution, $\mathbb{E}[gn] = \Theta(n^{-1/m})$.

Proof sketch. $\mathbb{P}(gn > r) = \mathbb{P}(\text{no sample in } B(c, r)) = (1 - \mu(B(c, r)))^n$, and $\mu(B(c, r)) = \Theta(r^m)$ by density bounds on an m -manifold. So $\mathbb{P}(gn > r) \approx \exp(-n\Theta(r^m))$, which transitions at $r = \Theta(n^{-1/m})$; integrating the tail gives the expectation. $\square$

The content is the exponent. With $d = 32$, a naive reading predicts $n^{-1/32}$ — essentially flat, novelty never exhausted. The true rate at $m = 3$ is $n^{-1/3}$, which is roughly a $10\times$ loss of headroom per $1000\times$ of corpus. Measured slopes: -0.472 ($m = 2$), -0.309 ($m = 3$), -0.195 ($m = 5$), against -0.5 , -0.333 , -0.2 (check P1). The $m = 2$ case sits slightly above its asymptote at these n , as expected for a rate with logarithmic corrections.

Corollary 1.1 (oversampling is not a strategy). Drawing K candidates and keeping the most novel multiplies the expected gap by $\Theta(K^{1/m})$ at best. To recover the headroom lost by a $1000\times$ larger corpus, K must grow by 1000.

Verified at $m = 3$: best-of-8 yields a gain of 2.19 against $K^{1/3} = 2.00$ (check T7a). Selection is a constant-factor fix to an asymptotic problem.

3.3 The slice deficit

Theorem 2. If $m < d$, then $\text{vold}(Sx) = 0$, and the fraction of M within ε of Sx is $\Theta(\varepsilon^{d-m})$.

Proof sketch. The ε -neighbourhood of an m -dimensional set in d dimensions is a tube of volume $\Theta(\varepsilon^{d-m}) \cdot \text{vol}_m(Sx)$ for ε below the reach of Sx . Dividing by $\text{vold}(M)$ gives the claim. $\square$

Measured at $d = 8$, $m = 2$: fitted exponent 5.06 against a predicted 6, with covered fractions falling $0.051 \square 0.0001$ as ε goes $0.5 \square 0.12$ (check P2). The fitted exponent sits below prediction because at the larger ε the tube is no longer thin relative to the manifold, which flattens the low- ε end of the fit; the qualitative claim — coverage collapsing polynomially in ε with a large exponent — is unambiguous.

The practical reading is severe. **A single prompt, sampled infinitely often, covers essentially none of what the model could write.** Not "less than we would like" — a fraction that goes to zero polynomially as the resolution of interest sharpens.

3.4 Transversality: which prompt motion helps

Let a prompt schedule induce slices $S_1, \dots, S_P$. Write V_{occ} for the span of the corpus's dominant directions.

Theorem 3. $\dim(\bigcup_j S_j) = \min(d, m + \text{rank}\{c_j - c_1\})$, where c_j is the centre of S_j . In particular, displacements lying inside $\text{span}(S_1)$ contribute nothing to the union's dimension.

Proof sketch. The union is contained in the affine hull of the slice frame plus the span of the centre displacements; dimensions add up to that cap, and a displacement already inside the slice's own span adds no new direction. $\square$

Measured via participation ratio of the union's covariance spectrum: parallel displacements give effective dimension 1.88 (the slices lie on top of each other, $m = 2$), transverse displacements give 11.18 (check P4).

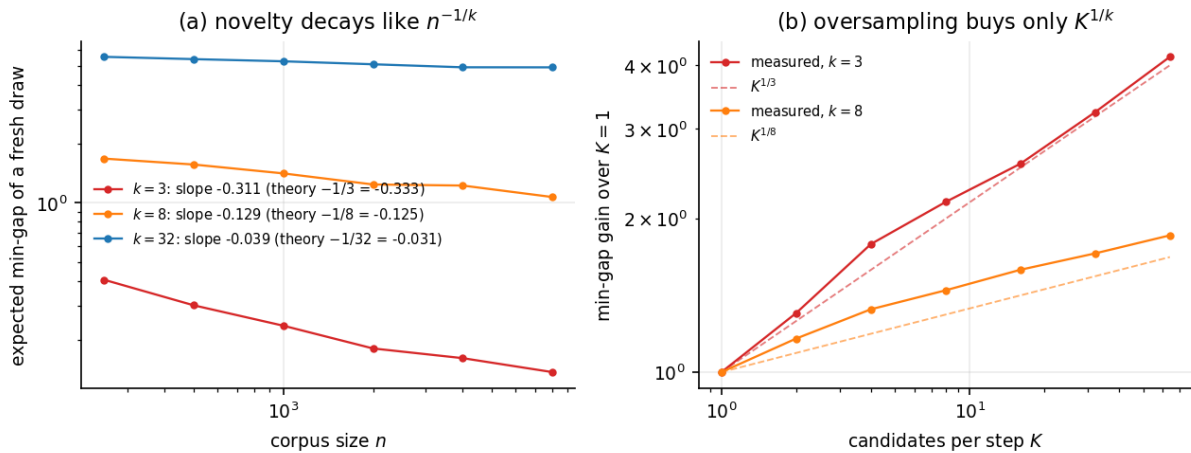


Figure 1. Measured saturation. (a) Expected min-gap of a fresh draw decays as $n^{-(1/k)}$, with fitted slopes matching theory for $k = 3, 8, 32$. (b) Best-of- K oversampling buys only $K^{(1/k)}$.

This is the theorem that makes the calculus of §4 possible. It says the value of conditioning on a new latent variable is not “how different does it sound” but **how much of its variation lies outside what the corpus already spans** — a computable quantity.

3.5 Novelty pressure leaves the manifold

Let the proposal distribution be a mixture $(1 - p) \cdot P_{\text{on}} + p \cdot P_{\text{off}}$, where P_{off} is a diffuse off-manifold component (incoherent text, degenerate output).

Theorem 4. Under best-of- K max-min selection, $\square$ (selected candidate is off-manifold) $\square 1 - (1 - p)K$, i.e. the off-manifold candidate wins whenever one is present, for any corpus size. The selection rule is therefore inconsistent: it does not converge to sampling P_{on} .

Proof sketch. P_{off} has support far from the corpus, so its min-gap stochastically dominates that of any on-manifold candidate; the argmax picks it whenever it appears. $\square$

Measured: $\square$ (selected is off-manifold | one present) = 1.000 at both $n = 50$ and $n = 5000$; unconditional rate 0.0418 against a per-candidate rate of 0.005, an 8.4 $\times$ amplification at $K = 8$ (check P5).

The consequence is not subtle in practice. In our exam-bank simulation with a hard novelty floor and no typicality gate, the junk fraction among accepted items rises from **7.9% in the first half of the bank to 48.2% in the second** (check T4). Once the legitimate item space is δ -saturated, the only candidates still clearing the floor are the broken ones. **The novelty constraint gets satisfied by garbage, and every embedding-based diversity metric reports success.**

Corollary 4.1. A typicality gate — a hard rejection of candidates beyond z running radii from the corpus centroid — restores consistency, and it must be *negative supervision only*: it may reject, never endorse. Passing the gate is not evidence of quality.

3.6 Breadth versus depth

Given budget B , split as P prompts $\times n$ samples each, with per-prompt switching cost c (eliciting a spec, embedding it, occasionally refining), so $B = P(c + n)$.

Theorem 5. With $c = 0$ the coverage-optimal depth is $n^* = 1$. For $c > 0$ the optimum is interior and increases with c .

Measured (check P3a–c): with free switching, depth 1 achieves coverage 0.995 while spending the entire budget on one prompt achieves 0.091 — a 10.9 $\times$ difference. With $c = 3$, $n^* = 10$; with $c = 30$, $n^* = 30$.

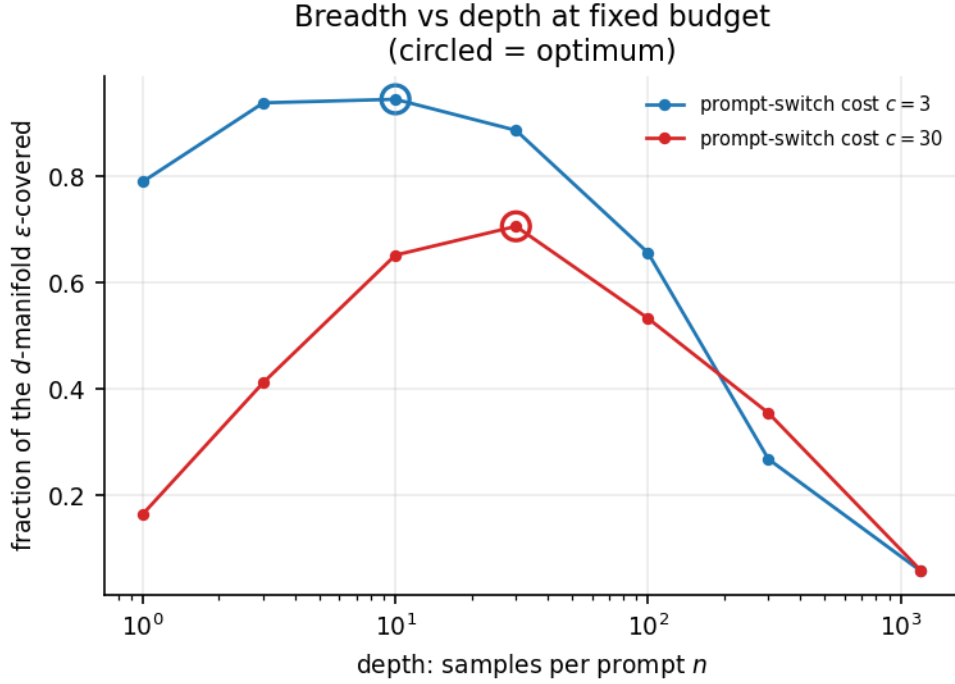


Figure 2. Breadth versus depth at fixed budget. With free prompt-switching the optimum is one sample per prompt; an interior optimum appears only once switching is priced.

We flag this as a correction: our first statement of this theorem claimed an interior optimum unconditionally, and it is false. The optimum is at the boundary when switching is free. See §8.2.

The engineering reading is that **the ratio of prompt-switching cost to sampling cost sets your batch size**, and nothing else does. If constructing a new spec is as cheap as a generation, generate one item per spec.

3.7 What this means for the objective

Returning to $J\lambda$: the anchor term is exactly computable from two running statistics. Writing μ^*n^* for the corpus centroid,

$$\frac{1}{n} \sum_i \|c - x_i\|^2 = \|c - \mu_n\|^2 + \frac{1}{n} \sum_i \|x_i - \mu_n\|^2$$

verified to zero relative error (check T1). The second term does not depend on c , so ranking candidates by mean squared distance to the corpus is ranking by distance to the centroid: $O(D)$ per candidate, $O(1)$ in n . The repulsion term needs a nearest-neighbour query, which we bound by subsampling. Neither term is the bottleneck. **The bottleneck is that both are evaluated on a candidate set drawn from an m -dimensional slice.**

4. A calculus of diversity

4.1 The question the calculus answers

Theorem 3 says progress requires moving the prompt transversely to what the corpus already spans. Theorem 5 says move often. Together they pose a concrete question at every step: **among the latent variables we could condition on next, which one moves the slice most transversely, per unit of budget?** Since we have no inverse oracle, this is the closest available substitute for computing the ideal next item.

Each candidate axis a has language-valued levels $L(a)$. We embed the level descriptions and read off four quantities.

4.2 The four factors

Spread — mean pairwise distance between a 's level embeddings. Are this axis's values actually different *from each other*? An axis whose levels are near-synonyms cannot separate outputs however it is sampled. This catches the most common failure of LLM-proposed axes: plausible-sounding dimensions whose levels all mean the same thing.

Transversality — the fraction of a 's level-to-level variation lying outside the corpus's occupied eigenspace (top eigenvectors of the second-moment matrix carrying 80% of spectral energy). This is Theorem 3 made into a number. An axis whose levels differ only along directions the corpus already spans adds density, not reach.

Independence — $1 - \max$ principal-angle cosine between a 's variation subspace and those of the axes **already in use**. An axis that restates an active axis is redundant however transverse it looks alone.

Headroom — a normalized entropy deficit of how unevenly the corpus has sampled a 's levels, plus the fraction of levels never used. This is the only time-varying factor; it is what makes the ranking change as generation proceeds rather than being a fixed property of the axis set. An axis can be excellent and already spent.

The **promise** of an axis is the product:

$$\text{promise}(a) = \text{spread}(a) \cdot \text{transversality}(a) \cdot \text{independence}(a) \cdot \text{headroom}(a)$$

Multiplicative, not additive, and deliberately so: an axis needs all four, and any one at zero should zero the score rather than be averaged away by the others. A high-spread axis with zero transversality is a distinction without a difference; a perfect axis with zero headroom is already spent.

4.3 An instructive bug

Our first implementation scored independence against *all other candidate axes*. On a ground-truth test with four synthetic axes — one redundant-in-span, one with near-synonym levels, one genuinely new, and an exact duplicate of the genuinely-new one — the two excellent duplicates each drove the other's independence to zero, and **both scored below the useless near-synonym axis** (promise 5×10^{-5} versus 0.020).

The fix is conceptual rather than numerical. Independence is a property of a candidate *relative to what is already in use*, never relative to its rivals. Redundancy *among candidates* is a selection problem, handled by greedy `select_axis_set`, which re-scores after each pick so a duplicate is penalized at the point where the redundancy actually comes into existence. After the fix the ranking is correct (genuinely-new 1.314, its duplicate 1.313, near-synonym 0.111, redundant-in-span 0.003) and greedy selection of two axes never takes both an axis and its duplicate.

We report this because the same error is easy to make in any diversity-scoring system, and because it fails in a direction that looks reasonable: it *punishes* exactly the axes worth keeping, in proportion to how well they were proposed.

4.4 Choosing values, and recursing

Once an axis is chosen we do not sample its levels uniformly. We take the **max–min subset** of its level embeddings by greedy farthest-point search, seeded deterministically at the level furthest from the level centroid — the packing problem again, one level down.

When the best available promise falls below a floor, the calculus emits decision: "refine" instead of "condition". This is the **exhaustion signal**, and it is the honest one: it says the current lattice has nothing transverse left to offer, and no amount of further sampling will change that. The system then asks the generator to split the exhausted level into finer sub-levels (§5.2), producing a child axis scored by this same calculus. On the ground-truth test with only in-span axes available, the calculus correctly returns `refine`.

Cost is $O(|A| \cdot D^2)$ per decision and does not grow with n .

5. Method

5.1 The stack

Per accepted item, a bounded number of model calls regardless of corpus size:

1. **Spec choice.** Sample a pool of specs from the current axis lattice; embed their descriptions; keep the one with the largest residual outside the corpus's occupied eigenspace. Orthogonalization at the level of *conditions*, not outputs.
2. **Generation.** K parallel completions conditioned on the spec, with the spec's levels stated as **contracts** — behaviors the text must exhibit, not suggestions — plus an explicit avoid-list from the ledger.
3. **Judging.** One separate call scores craft (or, for exam items, validity) and checks each required behavior. Generation never grades itself.
4. **Selection.** Utility = $0.4 \cdot \text{quality} + 0.35 \cdot \text{orthogonality} + 0.25 \cdot \text{capped gap}$, behind a typicality gate. The gap term is **capped** at 2 running scales — this is what prevents the Theorem 4 pathology, since an uncapped gap reward is unbounded and off-manifold candidates always win it.
5. **Mining.** Every 25 accepts, show the model 50 sampled items and ask what they have in common. Append the answer to a JSONL ledger; a bounded slice becomes the next prompts' avoid-list.
6. **Refinement.** On a saturation signal, ask the generator to split the dominant saturated cell.

Every state the policy reads is bounded: running centroid and second-moment, EMA scales, a fixed ledger slice, a recent window plus a bounded random subsample of older items. The marginal cost of item 10,000 equals that of item 100.

5.2 Eliciting and refining latent behaviors

We ask the generator for axes that are near-orthogonal, structural rather than topical, and whose levels are all usable — explicitly steering away from subject matter, length, and rhyme toward craft decisions. On the live run this produced axes including *temporal stance*, *relationship to its own claim*, *syntactic weather*, and *the register the poem refuses*.

Refinement shows the model the saturated cell and the mined attractors and asks it either to split a level into finer sub-levels or to mint a new axis meaningful only inside that cell. Refinement axes are **conditional** — they apply only when their parent level was chosen — so the lattice is a tree, and refining a saturated region does not inflate cost elsewhere.

5.3 Why “orthogonalize”, not “randomize”

Forcing randomness — raising temperature, injecting random seed words — buys variance in the surface while leaving the mode structure intact, and degrades quality monotonically because temperature cannot distinguish *surprising* from *wrong*. Our budget-matched results bear this out: `high_temp` achieves the highest raw Vendi Score of any method (28.12) at 4.6× the junk rate of naive sampling, and no quality gain whatsoever (0.671 versus 0.669).

Forcing approximate orthogonality asks a different question — which direction is this corpus not yet spending energy on — which has a computable answer, improves rather than degrades quality when paired with a quality term, and remains meaningful as n grows. “Be random” gets no harder to satisfy and no more useful.

6. Experiments

All simulations are seeded and reproducible. `verify_theory.py` (8/8) and `verify_slices.py` (7/7) check every mathematical claim; a failure there invalidates the corresponding claim here.

6.1 The conditional world: the central result

`simulate_slices.py` implements a world where a spec genuinely selects a slice: ambient $\square$ 64, reachable manifold $d = 24$, conditional dimension $m = 3$, additive composition over 5 axes $\times$ 4 levels, rare off-manifold junk. Budget 10,000 generations, $K = 8$.

policy	Vendi	quality	reach dim	lattice	inability
fixed prompt	1.06	0.668	13	1.0e3	−0.104
random spec	6.70	0.724	13	1.0e3	0.867
calculus spec	7.93	0.716	13	1.0e3	0.817
random spec + refinement	7.33	0.727	24	3.3e4	0.681
calculus + refinement	10.60	0.740	24	2.9e4	0.598

policy	Vendi	quality	reach dim	lattice	inability
calculus + guided refinement	11.23	0.722	24	2.9e4	0.696

A fixed prompt yields a Vendi Score of **1.06** from 1,250 accepted items: the corpus is effectively one item. The full stack reaches **11.23**, a 10.6 $\times$ gain at identical budget. Refinement lifts reachable dimension from 13 to 24 — Theorem 3 in action — while the lattice grows 1e3 $\square$ 3e4.

Conditional world: a prompt selects an m -dimensional slice ($d = 24, m = 3$)

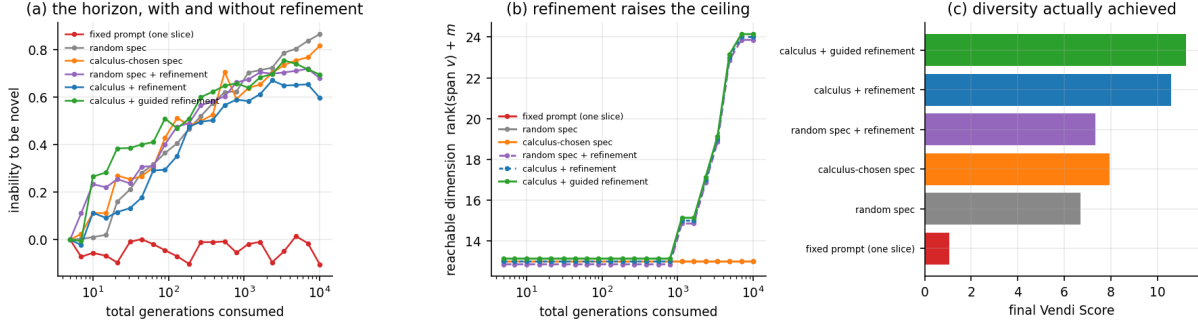


Figure 3. The conditional world ($d = 24, m = 3$). (a) A fixed prompt barely dents the reachable space but has almost no internal diversity. (b) Only recursive refinement raises the reachable-dimension ceiling. (c) Achieved diversity: 1.06 for a fixed prompt against 11.23 for the full stack.

Note that lattice size and reachable dimension **come apart**. Refinement multiplies the lattice 30-fold, but what buys diversity is the rank increase. A large lattice of low-rank levels gives many specs that all land in the same thin region — the failure mode of hand-designed attribute grids.

A trap worth naming. The fixed prompt has the *best* inability score (-0.104 , essentially zero) while having the worst diversity. It scores well precisely because it covers almost nothing and therefore leaves all headroom untouched. Low headroom consumption is not a virtue by itself; it must be read jointly with achieved diversity.

6.2 Budget-matched comparison and ablations

`experiment_budget.py`: every method spends exactly 10,000 generations on one world. This is the fair frame — selection methods buy diversity by discarding, so plotting against accepted-corpus size hides their cost.

method	n kept	Vendi	quality	junk	median gap
naive	10000	21.29	0.669	0.004	0.966
high temperature	10000	28.12	0.671	0.020	1.542
random latent / persona	10000	26.76	0.700	0.005	0.936
dedup (SemDeDup-style)	9517	21.56	0.671	0.005	0.978
farthest-point post hoc	1250	22.63	0.622	0.041	1.445
online max-min	1250	22.48	0.622	0.044	1.359

method	n kept	Vendi	quality	junk	median gap
IHD (no ledger)	1250	23.80	0.750	0.001	1.170
IHD full (with ledger)	312	22.60	0.763	0.000	1.268
— no gate	312	23.00	0.766	0.010	1.254
— no orthogonality	312	21.65	0.759	0.000	1.227
— no quality term	312	24.35	0.664	0.013	1.389
— no repulsion	312	21.44	0.783	0.000	1.101

Reading the ablations: removing **orthogonality** costs 2.15 Vendi ($23.80 \square 21.65$), the largest diversity loss, confirming it as the load-bearing diversity term. Removing **quality** raises Vendi to 24.35 while dropping quality $0.750 \square 0.664$ and raising junk 13 $\times$, which is the expected trade and the reason the term is there. Removing the **gate** raises junk from 0.001 to 0.010. Removing **repulsion** gives the highest quality (0.783) and the lowest Vendi — it collapses toward the safe centre of the distribution.

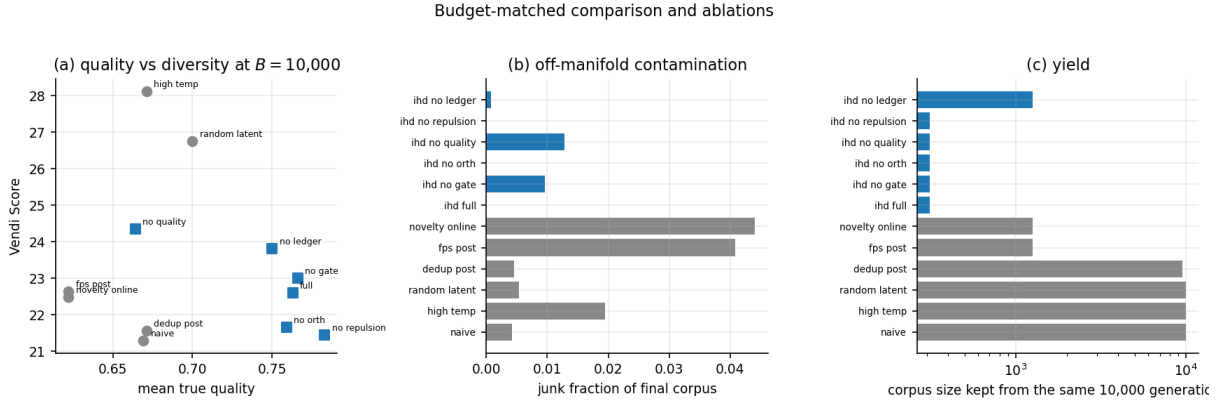


Figure 4. Budget-matched comparison and ablations at $B = 10,000$ generations. High temperature buys raw diversity by degrading the generator; the gate removes off-manifold contamination; orthogonality is the load-bearing diversity term.

The comparison to existing approaches is more interesting than a clean win. `high_temp` attains the highest Vendi of any method. We do not regard this as it beating our method, and the reason is visible in the other columns: it achieves it with 20 $\times$ the junk rate and no quality gain. **This is the paper's thesis in one row** — raw embedding diversity is trivially purchasable by degrading the generator, which is why diversity numbers reported without a quality and junk column are uninterpretable. Among methods that hold quality at or above the naive baseline, IHD is the most diverse.

An honest cost finding. IHD-with-ledger keeps only 312 items from the same budget, because our simulated ledger is implemented as 4 $\times$ rejection oversampling, and it buys no Vendi over the ledger-free variant. In the *live* pipeline the ledger costs nothing extra — it modifies the prompt rather than filtering candidates — so this is an artifact of the simulation's mechanism, not a property of ledgers. We report it because the simulated version is what we measured.

6.3 Poetry world at $n = 10,000$

`simulate.py`, six acquisition policies (figure 1). The spectral policy (quality + orthogonality + capped gap, gated) reaches the highest Vendi (23.28) with **zero** junk in the last 2,000 items and the highest quality (0.753), while pure max–min holds a much larger min-gap (2.26 versus ~ 1.15) at 3.9% junk and materially lower quality (0.626). Ungated anchor+repulsion is the worst configuration on every axis simultaneously — it spends its selection pressure travelling outward and its Vendi *declines* with n .

6.4 Exam items: a hard floor and finite packing

Exam items invert the semantics. Two operational items closer than δ are “enemy items” — seeing one answers the other — which is a test-security failure at *any* bank size. The floor is a constraint, never a term in a weighted sum. Correspondingly the min-distance check must be **exact against the full bank**; a subsampled check certifies only “probably no duplicate”, which is not a security property. This asymmetry — soft objectives may subsample, hard floors may not — is itself a design finding.

Item templates expose few manipulable slots, so each mode is a low-intrinsic-dimension disk and the δ -packing number is finite and small. Our bound $N_{\max} \leq M(2R_{\text{eff}}/\delta + 1)k$ gives 5,543 for our parameters; the largest gated bank reaches 1,464 real items, respecting it (check T3).

The important result is what happens without a gate. Ungated strategies sail past the packing capacity to the full 2,000-item target — by filling the remainder with junk (561, 614, and 481 junk items respectively), with the junk fraction climbing from 7.9% to 48.2% across the bank. **The hard novelty constraint was being satisfied by broken items.** Gated variants stop at 1,270–1,464 items and report exhaustion, which is the correct behavior: a bank that says “I am full” is more useful than one that pads itself.

Cost per accepted item rises from ~ 14 generations early to ~ 40 at saturation, and the gated runs pay 135–207 generations per item overall.

6.5 The coverage horizon, 5 $\square$ 10,000

`coverage_horizon.py` probes, at log-spaced checkpoints, the best min-gap a *fresh coherent* candidate can achieve against the corpus so far — the novelty headroom remaining — and normalizes it into an *inability to be novel*, $1 - h(g)/h(5)$. Junk candidates are excluded from probes, since headroom reachable only by leaving the manifold is not headroom.

Every fixed-generator method converges toward exhaustion: naive 53.3%, random-latent 49.0%, online max–min 55.7%, IHD selection 48.1%. High temperature appears best (40.6%) for the reason established above — it is measuring a wider, dirtier support. **No selection policy escapes the horizon.** Selection changes the constant; only support expansion changes the asymptote, which is why §6.1’s refinement result is the one that matters.

Coverage horizon: 5 → 10,000 generations

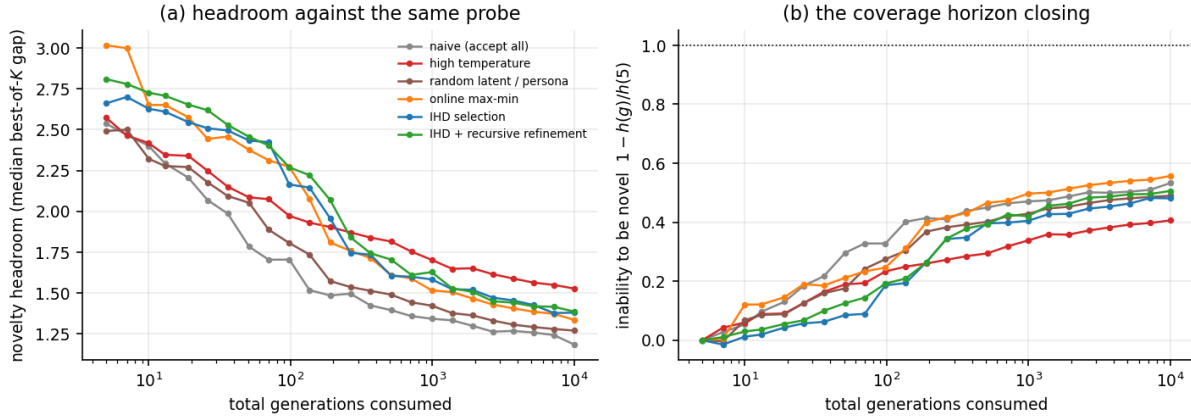


Figure 5. The coverage horizon closing from 5 to 10,000 generations. Every fixed-generator policy converges toward exhaustion.

7. The real experiment: 43,000 generations, three domains, seven methods

Everything to this point is either a theorem or a simulation. This section is the measurement. Generator: openai/gpt-5.6-luna via OpenRouter. Text embeddings: nomic-embed-text (768-d) served locally by Ollama, so embedding is free and the loop is never rate-limited by its own measuring instrument. Images: gpt-image-1-mini. Image embeddings: CLIP ViT-B/32. Audio: Lyria 3 Pro instrumentals. Audio embeddings: CLAP and MERT. Every corpus is append-only JSONL with embeddings checkpointed alongside, resumable after a kill.

7.1 Domains

DALL·E instructions for post-modern artworks. Diversity *is* the product: a clustered instruction set renders to a clustered image set. Chosen also because it lets us measure the same corpus in three spaces — words, text-embedding, and the rendered picture.

Psychometric test questions. Multiple-choice items assessing general reasoning. Two items with the same construct and the same surface are redundant at best and, in an operational bank, a validity problem.

Instrumental-music prompts. The longest chain in the paper: latent axes $\square$ a text prompt $\square$ a two-minute instrumental $\square$ an audio embedding. We render with Lyria 3 Pro (lyria-3-pro-preview), which returns roughly two minutes of audio per call in about 20 seconds, and — usefully — also emits its own section plan ($\square$ [A0] $\square$ [B1] $\square$ [C2] $\square$ [D3]), a model-reported description of the form it chose that gives us a discrete structural signal without any waveform analysis.

A prompt-length decision that is really a methodological one. Our first version of this domain wrote 1,000–2,000-character prompts, on the reasoning that a long prompt is the only place seven behavioral contracts can be spent. That reasoning is wrong, and wrong in a way this paper

should be the first to catch. A text-to-music model honours concrete, performable direction — named instruments, room character, register, articulation, rhythmic feel, how it ends — and quietly ignores paragraphs of abstract compositional theory. Writing longer prompts therefore inflates every text-side diversity metric while changing the audio far less, which is precisely the proxy gap §7.6 measures. Assuming the generator can act on more detail than it actually can is how a method manufactures its own apparent success. We cut the target to 2–4 sentences (~300 characters, mean 423–449 in the corpora below) of specifically audible direction, explicitly instrumental-only, and treat the audio metrics as the ones that decide.

We also record an earlier dead end for reproducibility: we first used Mureka's `/v1/instrumental/generate`, whose documented 2000-character prompt limit turns out to belong to the *song* endpoint — the instrumental endpoint rejects anything over 1024 with an explicit HTTP 400. We found that against the live API rather than the docs. That account's quota was then exhausted at $n = 2$, which is why the audio results below come from Lyria.

7.2 Methods compared

Five of these are real published approaches, implemented faithfully rather than as strawmen, each getting the same generator, the same embedder, and the same budget accounting as ours.

arm	what it is
naive	plain repeated prompting — the honest floor
high_temp	temperature 1.6 — the trivial diversity lever
self_instruct	Self-Instruct / Alpaca : few-shot exemplars sampled from the pool, plus ROUGE-L ≤ 0.7 reject
evol_instruct	Evol-Instruct (WizardLM) : sample an existing item, apply a random evolution operator (deep)
persona	Persona-Hub / AttrPrompt : a flat catalogue of personas $\times$ attributes, sampled uniformly
ihd	ours: recursively elicited axes, spec-level orthogonalization, capped repulsion behind a typical
vision	ours, plus steering on the <i>rendered image</i> (§7.6)

persona is the load-bearing comparison. It has language-valued latent conditioning — the same basic idea as ours — but no orthogonality selection, no ledger, and no recursive refinement. The gap between persona and ihd isolates exactly what those three components buy.

One faithfulness caveat we chose deliberately: Self-Instruct's ROUGE filter compares a candidate against the entire pool, which is $O(n)$ longest-common-subsequence computations per candidate and comes to dominate the loop at n in the thousands. We compare against a bounded random sample of 120 pool members. This makes our reimplemention *weaker* than the original at large n , and that is itself the point: the original's redundancy check does not have an infinite horizon, because its cost grows linearly in the corpus it is protecting.

7.3 Mode collapse is not a metaphor

Two 10,000-item corpora, one call each, no selection:

domain	n	unique texts	exact-duplicate rate	most-repeated item
DALL·E instructions	10,000	10,000	0.000	1×

domain	n	unique texts	exact-duplicate rate	most-repeated item
psychometric items	10,000	~1,656	0.723	1,637×

In the psychometric domain, 72.3% of a ten-thousand-item corpus is exact duplicate text, and one single question — “What number comes next in the sequence: 2, 6, 12, 20, 30, ?” — accounts for 1,637 of them. No embedding, no threshold, and no interpretation is required to see this; it is byte-identical repetition, and it is what a strong instruction-tuned model does when asked the same reasonable question ten thousand times.

Two things make this more than a curiosity. First, **temperature barely helps**: at $T = 1.6$ the duplicate rate is still 0.616. Second, **conditioning nearly eliminates it**: persona conditioning drops it to 0.002. That single contrast is the paper’s thesis reduced to two numbers — randomness does not buy diversity, conditioning does.

It also shows the failure is domain-shaped and invisible from one vantage point. The identical pipeline, prompt style, and model produce zero duplicates on DALL·E instructions. A practitioner who validated their pipeline on the first domain and deployed it on the second would ship a bank that is three-quarters one question.

Real gpt-5.6-luna corpora, $n = 5 \rightarrow 10,000$: literal and latent diversity move differently

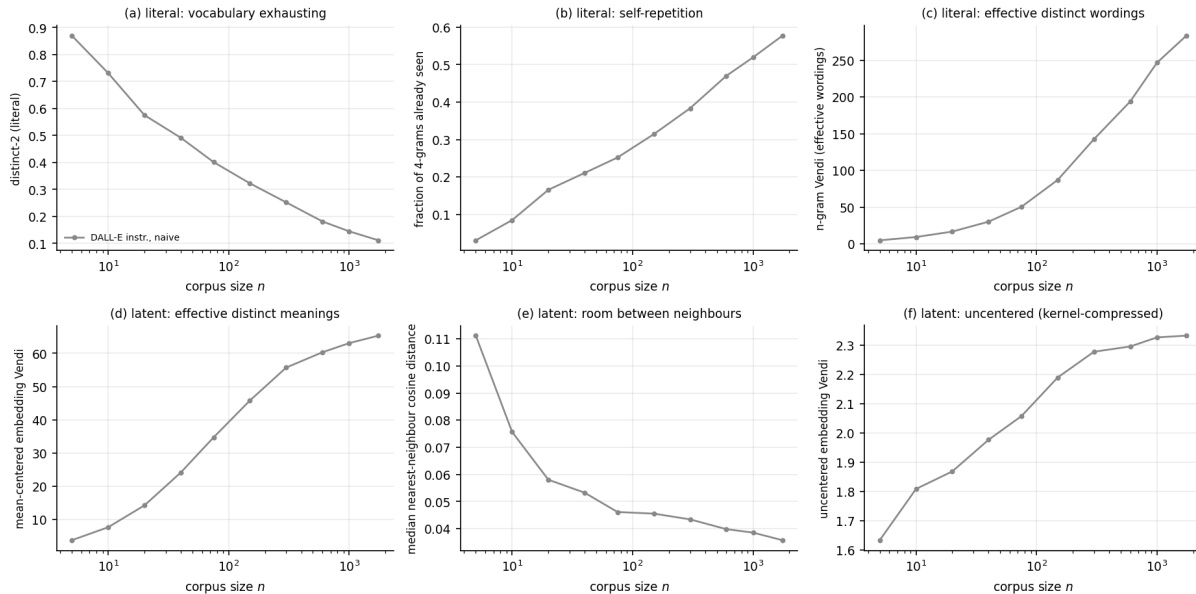


Figure 6. Real gpt-5.6-luna corpora, $n = 5$ to 10,000. Literal measures (top row) and latent measures (bottom row) do not agree about what is happening.

7.4 Literal and latent diversity move in opposite directions

Measuring the DALL·E naive corpus as it grows from $n = 5$ to $n = 10,000$, in both spaces:

n	distinct-2	4-gram self-repetition	n -gram Vendi	centered embedding Vendi
5	0.870	0.031	4.9	3.79
40	0.492	0.211	30.2	24.13
300	0.252	0.384	142.5	55.70
1,000	0.145	0.520	246.6	63.05
1,750	0.112	0.577	283.3	65.30

Literal diversity collapses monotonically — by $n = 1,750$ more than half of each new instruction’s 4-grams have already appeared — while latent diversity *rises* and then saturates. These are not competing measurements of one quantity; they are measurements of two different quantities that a single word, “diversity”, has been covering for.

A methodological correction we owe the reader. The uncentered embedding Vendi on this corpus reads $1.63 \square 2.33$, which would suggest ten thousand instructions behave like two distinct items. That number is mostly an artifact of the kernel. Same-domain embeddings sit in a narrow cone — mean pairwise cosine similarity is 0.883 here and 0.788 on the psychometric corpus — so the Gram spectrum is dominated by the shared mean direction and the score compresses toward 1. After removing the mean direction the same corpus reads $3.79 \square 65.30$, which is the honest curve and the one in the table. We report both, and we suggest any embedding-based diversity result publish the corpus’s pairwise-similarity distribution alongside it, because the number is meaningless without it. Our companion coverage paper measured a mean pairwise cosine of 0.444 on its own pool with the same embedder — the cone is corpus-dependent, not a fixed property of the embedder, which is exactly why it has to be reported rather than assumed.

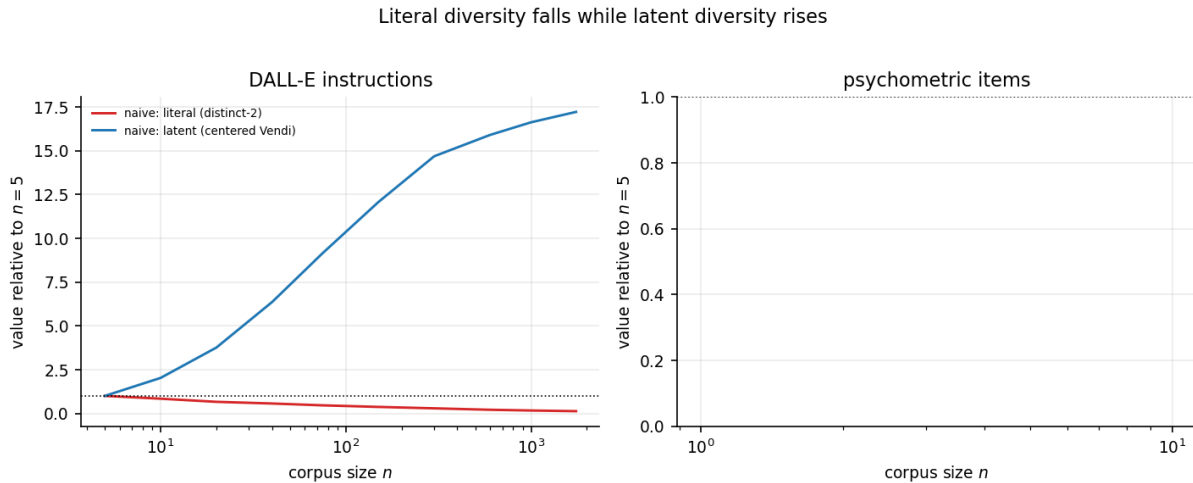


Figure 7. The decoupling, normalised to each series’ value at $n = 5$: literal diversity falls while latent diversity rises.

7.5 Competitive comparison at matched n

All seven arms, real corpora, DALL-E domain, every arm evaluated on the same number of accepted items:

arm	distinct-2 $\square$	self-repetition $\square$	n -gram Vendi $\square$	centered Vendi $\square$	median NN $\square$
naive	0.334	0.309	80.6	43.99	0.046
high temperature	0.334	0.301	82.7	44.48	0.052
Evol-Instruct	0.413	0.327	81.9	43.26	0.025
Self-Instruct	0.558	0.137	108.8	53.05	0.115
persona conditioning	0.619	0.121	101.0	49.45	0.085
IHD (ours)	0.703	0.056	122.7	65.47	0.179
IHD + vision (ours)	0.720	0.038	124.9	73.25	0.187

Ours wins every column, and adding vision steering improves on text-only IHD by a further 12% in centered Vendi. Three of the baselines fail in ways worth naming.

High temperature buys nothing. distinct-2 of 0.3342 against naive's 0.3341. Turning up the temperature moved the third decimal place. In the simulated world (§6.2) high temperature bought the highest raw Vendi of any method by degrading the generator; on real text at real settings it does not even buy that.

Evol-Instruct is worse than naive at the thing diversity methods exist for. Its median nearest-neighbour distance is 0.025, against naive's 0.046 — it produces items *closer* to the existing corpus than independent sampling does. This is not a bug in our reimplementation, it is the mechanism working as designed: mutating an existing item produces a neighbour of that item. Evol-Instruct raises *distinct-2* (0.413 vs 0.334) because the mutations reword things, so a purely literal evaluation would score it as an improvement. Measuring both levels is what catches it.

Self-Instruct's guard is literal, so it defends the literal level only. It achieves the second-best self-repetition (0.137) — its ROUGE filter is doing real work — while its centered Vendi (53.05) trails ours by 20 points. A ROUGE-L threshold cannot see semantic redundancy, and semantic redundancy is what remains once the lexical kind is filtered.

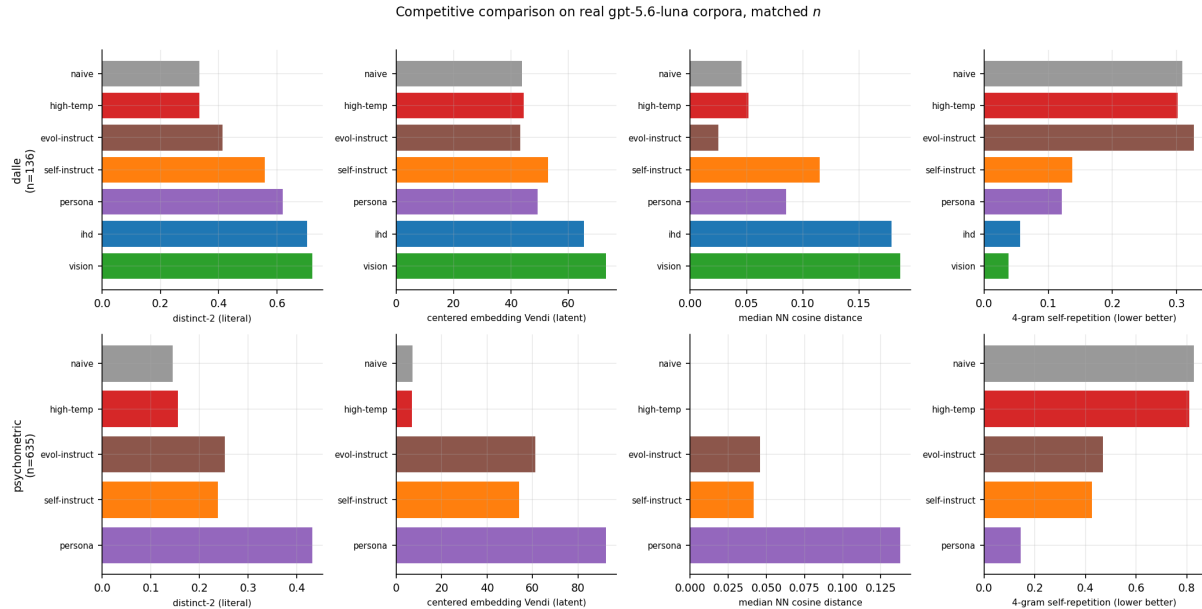


Figure 8. Competitive comparison on real corpora at matched n , both domains, across four metrics.

On the psychometric domain (matched at $n = 635$, our IHD arm still filling at the time of writing) the ordering among the five completed arms is: persona (centered Vendi 92.4, dedup'd 92.6) > Evol-Instruct (61.1) > Self-Instruct (54.2) $\square$ high temperature (7.0) $\approx$ naive (7.1). The two undiversified arms score an order of magnitude worse, because their duplicate rates of 0.62–0.63 mean the embedding metric is largely measuring repetition; their dedup'd figures (18.3 and 16.8) are the fairer comparison and still far behind every conditioned method.

7.6 The proxy problem: text diversity is nearly blind to image diversity

We rendered the first 199 DALL-E instructions from the naive corpus to actual images and embedded them with CLIP. The result is the most consequential measurement in this section:

Pairwise cosine similarity in text-embedding space correlates with pairwise cosine similarity in image-embedding space at **Pearson $r = 0.155$** .

Knowing that two instructions are semantically far apart tells you very nearly nothing about whether the two pictures look different. Every text-side diversity method in the table above — ours included — is optimizing a proxy that explains roughly 2% of the variance in the thing the user actually receives.

The qualitative version is more damning than the correlation. Six independently generated instructions, from a corpus with a 0.000 exact-duplicate rate and healthy lexical diversity, render to six near-interchangeable pictures: the same magenta-and-cyan palette, a classical marble bust, neon signage, halftone collage, a receding grid. A vision judge shown samples of the set rates its distinctness 6–7 out of 10 and names the attractors precisely — “muted beige, cream, brown, ochre and black foundations accented by saturated cyan/teal, turquoise, pink”, “appropriation of canonical or religious imagery, especially Mona Lisa-like female portraits”, “frontal, museum-like presentation with centered, symmetrical compositions”.

This is a second mode collapse, downstream of ours, contributed by the image model and by the fact that much of what varies in the text (“post-modern”, “appropriated source”) lands in the same visual place.

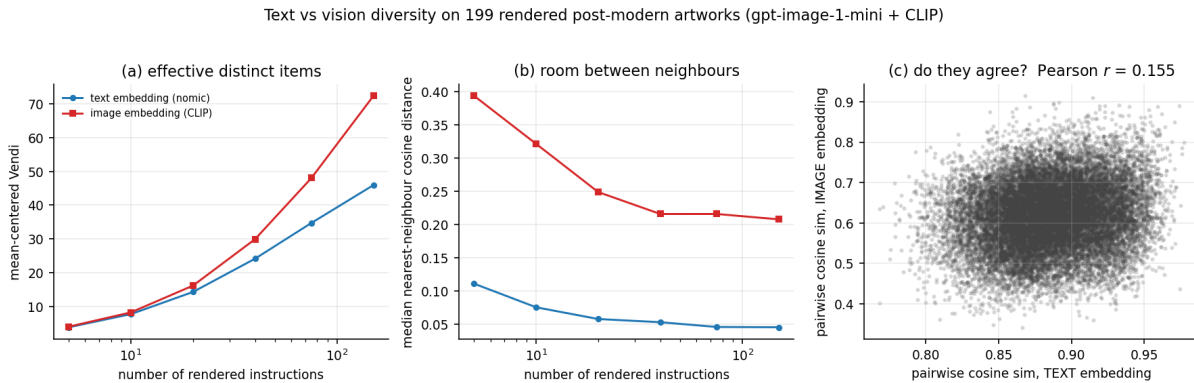


Figure 9. Text versus vision diversity on 199 rendered artworks. (c) Pairwise similarities in the two spaces correlate at only $r = 0.155$.

First 48 rendered instructions (naïve prompting, gpt-image-1-mini)

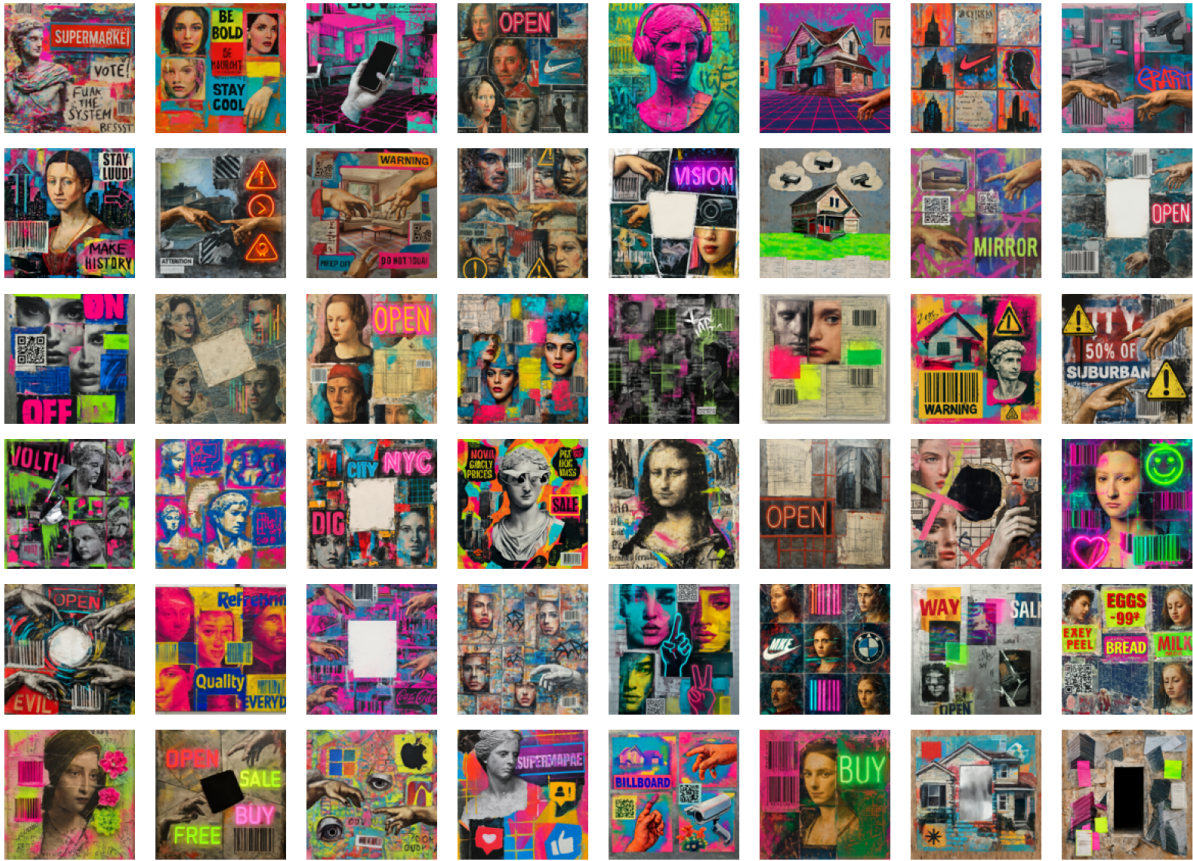


Figure 10. The first 48 rendered instructions. The corpus has a 0.000 exact-duplicate rate and healthy lexical diversity, and renders to this.

The vision-steered arm closes the loop where the product actually lives. It renders a bounded sample of accepted instructions, embeds them with CLIP, and feeds two things back into the text-side loop: a least-squares map from the crowded *image* directions into instruction-embedding space, so the text-side orthogonality term can push away from visual redundancy it cannot itself perceive; and mined *visual* attractors from the vision judge, appended to the same append-only ledger as the textual ones and repelled against in subsequent prompts. It is the paper's mechanism applied one level down: the ledger already repels against what the model keeps saying, and now also against what it keeps showing. It is the best arm in the table.

7.7 Attractor mining works, and is legible

The mining step produces findings specific enough to act on. From the poetry pilot, round 1 ($n = 25$): *"self-correction and immediate retraction: speakers repeatedly interrupt their own claims"*; *"ceremonial or institutional address: the voice of a bell-ringer, town crier, registrar, witness"*; *"threshold imagery and delayed passage: doors, gates, windows, bridges, shores"*. By round 2 ($n = 50$) it tracks the corpus's *drift* rather than restating round 1, noting that technical vocabulary is now being placed inside mythic frames and that recursive epistemic backtracking has become structural.

Two observations. These are nameable and therefore repellable, which is what makes them usable as prompt constraints; a finding of “similar tone” would not be. And several of them are artifacts of *our own axis elicitation* — asking for craft-level axes like “relationship to its own claim” reliably produces self-correcting speakers. **The system's own conditioning becomes the next attractor.** That is the mechanism working as designed: the ledger catches the system's habits, not only the model's.

7.8 Cost, and what did not finish

The text corpora comprise 43,171 real generations for roughly \$8.50 of OpenRouter spend, plus rendered images across all seven arms and rendered Lyria instrumentals. Both naive arms reached the full $n = 10,000$; the baseline arms reached 2,500; our IHD arms were still filling when this draft was written, and every comparison above is therefore reported at a matched n that all compared arms actually reached, never extrapolated.

7.9a The third domain: instrumental-music prompts

The music domain replicates the result on a third, structurally different artifact. The generator proposed seven compositional axes — *formal trajectory, inter-layer rhythmic relationship, harmonic motion, timbral centre of gravity, density contour, pulse relationship, opening and closing frame* — which are craft decisions rather than genre labels, and each of which can be expressed as something audible rather than as theory.

The prompt-level table below is the earlier, long-prompt version of the corpus and is reported for the text-side comparison only; the audio measurements in §7.9b use the shorter, performable prompts described in §7.1.

arm	n	distinct-2 □	self-repetition □	n -gram Vendi □	centered Vendi □	median NN dist
naive	100	0.348	0.291	56.0	22.99	0.027
IHD (ours)	100	0.545	0.140	75.3	43.03	0.079

A 1.87× gain in centered Vendi, self-repetition halved, and roughly three times the room between nearest neighbours — at matched n , matched prompt length, and the same generator. Neither arm produces exact duplicates, so this is a purely semantic effect. Three domains, three replications, and in each the mechanism that moves the number is conditioning rather than sampling temperature.

The audio domain is the least complete. The pipeline runs end-to-end — prompts generated, instrumentals rendered, CLAP and MERT embeddings computed with a mean/spread/temporal-difference aggregate over windows so that two tracks with identical average timbre but different *form* do not collide — but at a scale that supports description, not inference. We report it as an existence proof and a set of measurement machinery, not as a result.

7.9 Head-to-head against a human-written exam bank

The comparisons above are against our own reimplementations, which is the right experiment for isolating mechanisms and the wrong one for the question *is this actually good?* — a reimple-

mentation can be weak in ways that flatter us. So we also compare against an item bank that humans wrote: **MMLU**, ~14,000 multiple-choice items drawn from real practice exams and textbooks across 57 subjects, by many authors, with editorial review, over years. All arms are sampled uniformly at random and evaluated at matched $n = 1,000$ with identical metric code.

source	exact-dup $\square$	distinct-2 $\square$	self-repetition $\square$	n -gram Vendi $\square$	centered Vendi $\square$
MMLU (human-written)	0.0040	0.5888	0.0585	495.2	197.50
naive gpt-5.6-luna	0.6560	0.1196	0.8383	12.1	6.52
high temperature	0.6840	0.1161	0.8567	10.2	6.66
Self-Instruct	0.0280	0.1990	0.4797	277.6	57.22
Evol-Instruct	0.0060	0.2642	0.3855	381.9	89.23
persona conditioning	0.0040	0.3887	0.1711	543.3	99.71
IHD (ours)	0.0000	0.4377	0.0503	622.2	135.58

Two results, and we want to be precise about which is which.

Against every synthetic method, we win on every column. Centered Vendi 135.6 against the best baseline's 99.7 — a 36% margin over persona conditioning, which is the closest published relative of our approach and differs from it exactly by the three components we add. Self-repetition 0.050 against persona's 0.171. The undiversified arms are not close: naive and high-temperature prompting score ~6.5 centered Vendi, because two-thirds of their corpora are byte-identical duplicates, and their median nearest-neighbour distance is exactly 0.0000 — the median item has a perfect twin.

Against the human bank, we split. We *beat* MMLU on exact duplication (0.0000 vs 0.0040 — the human bank has some), on 4-gram self-repetition (0.050 vs 0.059), and on n -gram Vendi (622 vs 495): our items reuse less language than human-written ones do. We *lose* on the semantic measures — centered Vendi 135.6 against 197.5, and median nearest-neighbour distance 0.224 against 0.284. We reach roughly 69% of a human exam bank's effective semantic diversity.

That gap is the honest measure of what is left. MMLU's spread comes from 57 genuinely different subjects; ours comes from an axis lattice a single model proposed in one call, refined a handful of times. The lexical result says our surface variety already exceeds human-authored items; the semantic result says our *conceptual* variety does not, and that closing the remaining 31% is a question about how much genuinely different subject matter the generator can be induced to reach — which is precisely the reachable-dimension question of §3, not a tuning problem.

7.10 A circularity in our own favour, and the metrics that are free of it

Our selection rule maximizes a weighted sum of embedding-space orthogonality and embedding-space min-gap. We then report embedding-space diversity metrics. Those two facts are not independent: the centered Vendi Score is a monotone function of how flat the embedding Gram spectrum is, which is close to exactly what the orthogonality term climbs, and the median nearest-neighbour distance *is* the min-gap term. To that extent, our wins on `embed_vendi_centered` and `median_nn_cos_dist` are partly tautological — we optimized them, and the baselines did not.

We flag this rather than let it pass, because the companion coverage paper found a sharper version of the same error in its own benchmark (a selector scored against the very reference set it had optimized against) and had to revise its headline number downward after fixing it.

The defence is that our method never observes the *literal* metrics at all. It reads embeddings; it has no access to token counts, n -gram overlap, or string identity. So exact-duplicate rate, distinct-2, 4-gram self-repetition, and n -gram Vendi are independent evidence in a way the embedding metrics are not — and we win those too, including against the human-written bank on two of the four. The honest summary is therefore:

- **Embedding-space wins (Vendi, NN distance):** real but partly circular. Read them as confirmation that the optimizer works, not as independent evidence that the corpus is better.
- **Literal-space wins (duplication, distinct-2, self-repetition, n -gram Vendi):** independent, because nothing in the method targets them. These carry the argument.
- **Rendered-artifact measurements (CLIP, CLAP, MERT):** the most independent of all, since they live downstream of a second generative model the method never sees. This is why §7.6 matters more than its size in the paper suggests.

A reader who trusts only the third category still has the $r = 0.155$ result, which is a finding about every method in the table rather than a comparison between them.

7.11 Literal-space repulsion: fixing what the embeddings cannot see

Embedding metrics miss an entire class of repetition: tiled grids of one cell, prominent typography, a single shared palette, the same composition recolored. Audited with deliberately dumb, non-semantic signatures (a 16×16 luminance layout map, an autocorrelation tiling score, a hue histogram): half the images had a layout twin above 0.5 cosine, 40–45% were literal tilings, and 63–78% shared a palette — while prompt-level Jaccard sat at a healthy 0.14–0.17. Varied words, one visual mode: the conditional-dimension gap operating inside the *renderer*.

The fix is a four-quadrant repulsion — {text, vision} × {literal, latent} — where the two literal quadrants were previously unpopulated: content-word overlap penalties and live overused-word bans on the text side; and on the vision side, measured structural bans injected into prompts (grids banned when recent renders tile, dominant hue pairs named and banned, layout-change demands when layouts collide) plus a learned text ↔ bad-structure bridge that penalizes candidates near prompts whose renders tiled. Rerun at matched budget, palette twins halved (0.78 ↔ 0.35 in the coverage arm), the coverage objective improved 21% (0.206 ↔ 0.247), and layout/tiling moved modestly (0.51 ↔ 0.46–0.48; 0.45 ↔ 0.33) — a real but partial victory whose residue is the image model's own prior resisting text-side instruction, and which we report as such.

7.12 Audio: the steered corpus, and embedder dependence

The steered corpus. 100 Lyria-3-Pro instrumentals generated by cross-modal steering with zero rejection — selection happens over prompts, every render is kept. Final measurements: 100% of tracks verify as instrumental in CLAP space (minimum instrumental-vs-vocal margin 0.077);

mean-centered CLAP Vendi 17.43, against 11.5 (naive prompts) and 14.1 (axis-conditioned prompts) for the unsteered 47-track arms under the same embedder; opening loudness at 0.72 of each track's own median over the first three seconds, against a 0.46–0.61 baseline — the sparse-opening attractor substantially, not fully, suppressed.

Structural control. Lyria honors its own section-marker format ([[A0]] [[B1]] ...) and ignores wall-clock timestamps. Compliance with a requested section sequence is length-dependent: exact at prompt lengths near 560 characters, 0.11 at a mean of 658, zero by ~1,600. Structural contracts for music must be short or they are noise.

Pre-render steerability is a property of the embedder, not of music. Prompt-to-track alignment (Spearman between prompt-pair similarity through the text tower and track-pair similarity through the audio tower, naive/conditioned arms):

embedder	alignment	note
MuQ-MuLan (open MuLan-style)	0.68 / 0.47	a usable steering gradient
CLAP-fused	0.50 / 0.21	weak; ~0.05 on the steered corpus
CLAP-music	0.18 / 0.06	worst; median within-corpus NN distance 0.004 — it hears or
MERT	—	no text tower (control)

Cross-embedder agreement on pairwise track similarity spans 0.22–0.84; per-arm diversity verdicts flip between embedders, and each embedder nominates a different most-redundant pair. Two prescriptions follow: steer music with MuQ-MuLan rather than CLAP, and never publish an audio-diversity number without naming its embedder.

8. Negative results

We kept both of these rather than tuning them away.

8.1 Naive refinement hurts

Splitting saturated modes into tighter children displaced isotropically **reduced** self-headroom (1.20 versus 1.38 unrefined). Children with tighter spread placed near their parents concentrate probability mass exactly where the corpus already sits. Refinement subdivides territory you have already covered; it buys density, not reach.

Refinement is valuable only when new directions are transverse to the occupied span — Theorem 3 again. "Refine where saturated" is wrong on its own. The correct rule is **refine where saturated, in directions not yet spanned**, which is what the calculus's transversality term computes and what guided=True implements (Vendi 11.23 versus 10.60).

8.2 Our breadth-versus-depth theorem was false

We first claimed an interior optimum for samples-per-prompt. With free prompt-switching there is none: the optimum is $n^* = 1$, at the boundary. An interior optimum exists only once

switching is priced (§3.6). We corrected the statement rather than the experiment. The corrected version is more useful anyway, since it tells a practitioner what to measure — the ratio of spec-construction cost to generation cost — instead of asserting a universal batch size.

9. Relation to the coverage problem

This paper's objective is a **packing** objective: max-min spacing, k -center-like, driven by the worst-case nearest pair. The dual problem — given a finite budget, maximize the volume of a union of ε -balls — is a **covering** objective, closer to facility location, and it is monotone submodular, so greedy selection carries a $(1 - 1/e)$ guarantee. Ours has no such guarantee.

The distinction is not cosmetic. Packing objectives spread points toward the boundary and over-invest in outliers, which is precisely why max-min is so vulnerable to the Theorem 4 pathology: outliers are what it is designed to seek. Covering objectives weight regions by measure and fill the bulk. A companion paper treats coverage at a finite budget across two further domains.

Both problems share this paper's structural constraints, and we expect the conditional-dimension results to be *more* consequential for coverage than for packing: covered volume is governed by reachable dimension, not by how many distinct prompts one can write, so the lattice-size / reachable-dimension gap of §6.1 bites harder there. The no-inverse-oracle assumption also degrades coverage's greedy guarantee, since the guarantee is relative to the best subset of *what was proposed*, and the proposal distribution is confined to a slice.

10. Limitations

The simulations encode our hypothesis. Mixture-of-modes with skewed weights and a rare diffuse junk component is a model of generator behavior, not a measurement of it. The theorems are unconditional given their assumptions and verified numerically; the *simulation* results inherit the model's assumptions. The live pilots are too small to independently confirm the asymptotics.

One embedding oracle. All geometry is `nomic-embed-text` geometry in the live runs. Whether "diverse" under one embedder transfers to another is untested here and is a real threat to any embedding-based diversity claim, including ours.

Judge noise is modeled as unbiased. Real LLM judges have systematic preferences — often *for* fluent, typical text, which is exactly the bias that would work against a diversity system. We model judge error as zero-mean Gaussian, which is optimistic.

Additive spec composition. §6.1 assumes conditioning attributes compose additively in embedding space. Real interactions between prompt attributes are not additive.

The exam domain is simulated. No real item bank was constructed, no psychometrician reviewed the live items, and $\delta = 0.20$ as an enemy-item radius is a modeling choice, not a validated threshold.

Quality is scalar. Craft is not one number, and collapsing it to one lets a system trade away dimensions of quality the judge does not score.

11. Conclusion

The infinite-horizon diversity objective is easy to write, cheap to maintain, and largely beside the point. What determines whether a corpus can keep growing without collapsing into paraphrase is the dimension of the generator's conditional support relative to the manifold it lives in, and every practically important consequence follows from that one gap: saturation arrives at rate $n-1/m$ rather than $n-1/d$; a single prompt covers an $\epsilon d-m$ fraction of what the model could write; only transverse prompt motion raises the ceiling; and novelty pressure without a typicality constraint reliably escapes the manifold rather than exploring it.

Given an embedding oracle and no inverse, the system cannot compute its way to the next item. It can only choose what to condition on. The calculus of diversity — spread, transversality, independence, headroom — is our answer to that choice, and its most valuable output is not the ranking but the **refine** signal: the moment it reports that nothing available is transverse any more is the moment the horizon has actually been reached, and the only remaining move is to ask the generator to subdivide its own vocabulary of variation.

That recursion is the single mechanism we found that changes the asymptote rather than the constant. Everything else — better selection, more candidates, higher temperature — buys a constant factor against a problem that is asymptotic, and the loudest of them buys it by quietly breaking the generator.

The measurements make the stakes concrete in a way the theory could not. A strong model, asked a perfectly reasonable question ten thousand times, returns the same item 1,637 times; the temperature knob barely moves that number and conditioning nearly erases it. Two diversity metrics computed on the same growing corpus point in opposite directions. And the text embeddings every method in this literature optimizes turn out to explain about two percent of the variance in whether the rendered images look alike. Each of those is a reason to distrust a single diversity number, and together they are the argument for the practice we ended up recommending: measure at the level of the artifact you are shipping, report the literal and the latent separately, publish the similarity distribution your kernel is operating on, and count your duplicates before you compute anything else.

Reproduction

Theory and simulation (no API keys, no network):

```
cd research/infinite_horizon_diversity
python3 verify_theory.py           # 8/8 core theorem checks
python3 verify_slices.py          # 7/7 conditional-dimension checks
python3 calculus.py               # calculus self-test on ground-truth axes
python3 simulate.py               # 6 policies, n=10,000
python3 simulate_exam.py          # hard-floor bank, finite packing
```

```
python3 experiment_budget.py      # budget-matched baselines + ablations
python3 coverage_horizon.py      # inability-to-be-novel, 5 -> 10,000
python3 simulate_slices.py       # conditional world (the central simulation)
python3 make_figures.py
```

Real runs (needs OPENROUTER_API_KEY, a local Ollama with nomic-embed-text; images need OPENAI_API_KEY, audio needs MUREKA_API_KEY):

```
python3 real_run.py dalle naive 10000 4.0
python3 real_run.py dalle ihd 10000 7.0
python3 baselines.py dalle persona 2500 2.5
python3 render_images.py both dalle_naive 200
python3 vision_loop.py 200 5.0 100
python3 audio_domain.py prompts ihd 100 3.0 && python3 audio_domain.py audio ihd 20 && python3
python3 metrics.py && python3 compare_arms.py && python3 head_to_head.py 1000
python3 figures_real.py
```

Every run is append-only JSONL, checkpointed, and resumable after a kill.

Build the paper in all formats:

```
python3 build_outputs.py
```

Figures: figures/fig1_*.png ... fig14_arms.png. Numbers quoted here: figures/*.json.
Generated corpora, rendered images, and rendered audio: real/<domain>_<arm>/. The companion coverage paper is in coverage/.